Letter of Transmittal

March 16, 2026

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Dear Reader,

I am delighted to submit this technical report on the topic “Using Artificial Intelligence and Algorithm Auditing to Reduce the Spread of Misinformation on Social Media Platforms.” The report focuses on the emerging problem of misinformation on social media platforms and how the use of artificial intelligence and algorithm auditing systems can help to mitigate the problem.

The purpose of the report is to examine the role of algorithm auditing systems in the proliferation of misinformation on social media platforms. The report also aims to evaluate the role of emerging technologies in reducing the proliferation of misinformation on the same platforms. By reviewing the current literature on the subject of misinformation on social media platforms, the report identifies the potential solutions to the problem. The report also examines the role of emerging technologies such as machine learning detection systems and algorithm auditing systems. The emerging technologies have the potential to improve the reliability of the information shared on the platforms.

I would like to take this opportunity to thank the reader for the opportunity to write on the subject. I would also like to take the opportunity to ask the reader to provide feedback on the report.

Sincerely,

Student

Artificial Intelligence and Algorithm Auditing to Reduce the Spread of Misinformation on Social Media Platforms

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Executive Summary:

The issue of misinformation on social media has become a serious problem on a global scale. The algorithms used on social media sites are based on maximizing user engagement. This has led to a situation where false or misleading information is promoted on these sites. This has contributed to the rapid dissemination of false or misleading information on digital platforms.

The purpose of this technical report is to explore how artificial intelligence and algorithm auditing can solve the issue of misinformation on social media sites. Current research has revealed that algorithm-based recommendation systems have contributed to the development of echo chambers. This is a digital environment where users are only exposed to information that is aligned with their personal views. This environment has contributed to the rapid dissemination of false or misleading information.

The report has reviewed various academic research on misinformation, social media algorithm systems, and machine learning-based detection systems. Research has revealed that machine learning systems can efficiently identify false or misleading information on digital platforms. Machine learning systems can identify patterns associated with false or misleading information.

Besides the use of technology in detecting misleading information, algorithm auditing technology also presents another possible solution to the problem. Algorithm auditing technology entails evaluating the behavior of recommendation system technology to ascertain its propensity to amplify misleading information and produce a biased information environment. Thus, implementing transparent algorithm auditing technology is likely to improve accountability and minimize algorithmic bias in social media platforms.

Despite the promising opportunities offered by the technology, it is also important to consider some of its possible shortcomings. For instance, the use of artificial intelligence technology in detecting misleading information might be limited by ambiguous information and satire. Moreover, its successful implementation is also dependent on the collaboration of technology companies and governments.

Artificial intelligence and algorithm auditing technology are important technological strategies aimed at improving the quality of shared online information.

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1. Introduction and Background

The rapid evolution of social media sites has revolutionized how we share and consume information. Social media sites such as Facebook, Twitter, and Reddit enable users to share information with global audiences in an instant. Although this has greatly enhanced access to information and has greatly improved the efficiency of communications, it has also led to an environment in which misinformation can spread instantaneously.

Misinformation is defined as the sharing of false or misleading information, irrespective of intent. On social media sites, this problem is particularly concerning due to its rapid spread. Unlike traditional media sites, which have some form of moderation in place to verify the accuracy of the information being spread, social media sites rely almost entirely on automated means of sharing and curating content.

The automated means of sharing and curating content are referred to as algorithms. These algorithms are built to maximize user engagement. Engagement-based algorithms favor those sites with the highest number of clicks, shares, comments, and emotional responses. Therefore, sensational, controversial, and emotionally stimulating content is often promoted much faster than factual but less engaging content. This is because it creates an environment in which misinformation can thrive.

It has been found that users tend to cluster in groups of like-minded people, also referred to as an echo chamber, in which they are only exposed to information that reinforces their existing beliefs (Cinelli et al., 2021). In such an environment, misinformation can spread much faster than in other environments without much opposition.

The consequences of misinformation, therefore, go beyond individuals. They have the potential to affect societal opinions, political processes, and even public health. Therefore, addressing misinformation is not only a technological but also a societal and ethical responsibility.

To address the issue of misinformation on social media, researchers have also proposed using artificial intelligence (AI) systems. Moreover, algorithm auditing is also suggested. This report aims to discuss how both of these solutions could be used in conjunction with each other to effectively address the problem of misinformation on social media platforms.

2. The Role of Social Media Algorithms in Misinformation

2.1 Algorithmic Content Distribution

Social media platforms have algorithms that decide what kind of content users should be exposed to. These algorithms work on user behavior, which includes likes, shares, watch time, etc. The basic idea behind using algorithms is to ensure that users are exposed to content that is aligned with their interests. The primary focus of all algorithms is to ensure that users engage

with the platform's content. Engagement is often used as a benchmark for the success of any social media platform.

Thus, with time, algorithms get extremely good at identifying what kind of content users are most likely to engage with. However, there is no focus on ensuring that users engage with true or accurate information. The focus is on ensuring that users engage with the content that is being displayed. Therefore, any form of emotionally stimulating or controversial content is likely to be preferred by algorithms, regardless of whether it is accurate or not.

This creates an environment where misinformation is more likely to succeed than accurate or true information.One of the most significant effects of this method of information distribution is that it leads to the creation of echo chambers. These are environments in which users are exposed to similar information that reinforces their already existing beliefs.

2.2 Echo Chambers and Reinforcement

Studies on social networks have revealed that users tend to cluster together in groups based on similar interests and opinions (Cinelli et al., 2021). In these clusters, information that is in line with the group's beliefs is likely to be accepted and spread, whereas information that is contrary to it is likely to be ignored or dismissed. This further reinforces the users' beliefs and opinions.

This leads to an increase in confirmation bias, making it difficult for users to critically think about information. Over time, this leads to a distorted view of reality in which misinformation is perceived as correct due to its similarity in nature.

2.3 Engagement-Driven Amplification

Studies have revealed that information that evokes emotions is disseminated faster compared to information that is neutral or factual in nature (McLoughlin & Brady, 2024). This is because information that evokes emotions such as anger, fear, and excitement is likely to be shared and engaged with. This is also due to the fact that algorithms amplify information that is likely to be engaged with. This is problematic as it leads to the amplification of misinformation. Figure 1 is presented to illustrate this point:

As seen in Figure 1, this demonstrates that misinformation is not only due to users' behavior but is also embedded in algorithms.

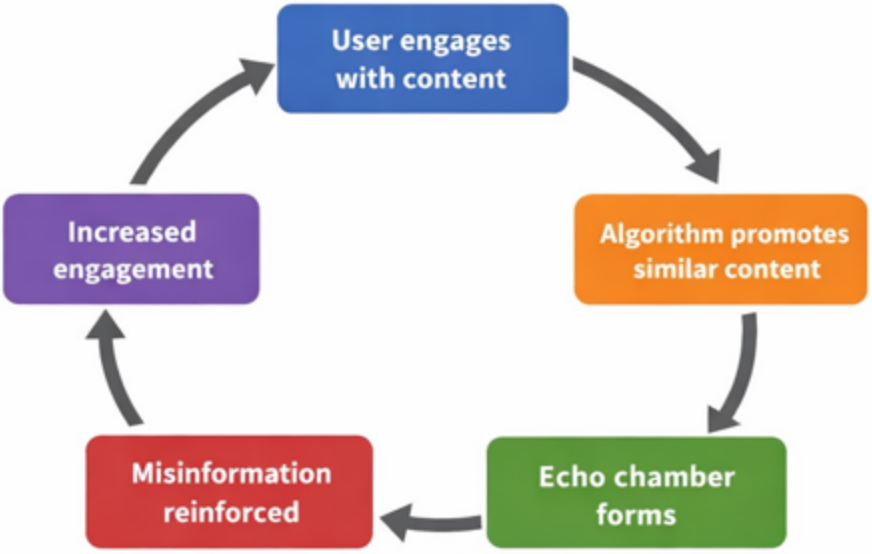

Figure 1: *Algorithmic Feedback Loop in Misinformation Spread*

3. Artificial Intelligence Technologies for Detecting Misinformation

Artificial intelligence has been recognized as an important tool in the fight against the spread of misinformation. AI has the ability to detect and identify misleading information. AI uses machine learning algorithms that analyze vast amounts of data and recognize patterns that indicate the presence of misinformation. AI algorithms can analyze text-based data and user interactions. Research indicates that AI algorithms can detect misinformation with high accuracy. AI algorithms analyze the linguistic features and structure of the content. AI algorithms also analyze the level of user interaction with the content. AI algorithms can detect the presence of misinformation with high accuracy.

One of the most important advantages of AI detection algorithms is that they are scalable. Social media platforms generate vast amounts of data every second. The volume of data generated on social media is impossible to manage. AI algorithms can continuously analyze the data and detect the presence of misinformation.

The detection algorithms developed by AI have several limitations. For instance, the detection algorithms may fail to distinguish between satire and misinformation. The algorithms may fail to

distinguish between opinion and misinformation. The algorithms may fail to distinguish between misleading and intended-to-be-misleading content.
Another limitation of AI detection algorithms is that they may generate false positives. The algorithms may detect the presence of misinformation in content that is actually true. The algorithms may generate false positives because of the way they analyze the content. The algorithms analyze the structure and the features of the content. The algorithms may fail to distinguish between intended-to-be-misleading and misleading content.

The above-mentioned limitations indicate that AI detection algorithms are reactive in nature. AI detection algorithms react to the presence of misinformation. AI detection algorithms react to the presence of misleading information. AI detection algorithms react to the presence of intended-to-be-misleading information. AI detection algorithms do not address the underlying mechanisms that enable the spread of misinformation.

4. Algorithm Auditing and Platform Transparency

Algorithm auditing is another way to mitigate misinformation, which is different from the above-discussed methods. Algorithm auditing is more focused on the dissemination of the content than the creation of the content itself.

Unlike AI detection systems, which are content-based, algorithm auditing is system-based. Algorithm auditing checks if the algorithms are giving more priority to the dissemination of false, biased, or damaging content.

Studies have shown that auditing frameworks are capable of detecting hidden biases in the recommendation system and understanding the impact of engagement-based models on the content (Chaudhuri et al., 2024). Algorithm auditing tests the algorithms under different conditions and checks if the algorithms are giving more priority to some types of content.
The first advantage of algorithm auditing is that it increases transparency in the algorithms.

Social media algorithms are considered “black boxes,” which means the internal decision-making process of the algorithms is not well understood. Algorithm auditing helps in understanding the internal decision-making process of the algorithms and the impact of the algorithms on the flow of the information.

Another advantage of algorithm auditing is that it helps in the development of more effective algorithms to reduce misinformation. Algorithm auditing helps in understanding the impact of the algorithms and how they are affecting the flow of the information.

The first disadvantage of algorithm auditing is that the access to the data of the social media platforms is limited, which makes it difficult for the researchers to perform the auditing (Chaudhuri et al., 2024). Algorithm auditing is considered a proactive way of mitigating misinformation.

5. Discussion

The analysis provided in this report clearly indicates that the problem of misinformation on social media is complex and multifaceted. The application of AI detection systems and the development of algorithm auditing frameworks address different aspects of the problem. The application of these two approaches in conjunction with each other provides a comprehensive solution to the problem.

The application of AI detection systems provides an effective solution to the problem of filtering out misleading information. The approach provides an immediate solution to the problem and can be highly effective in filtering out harmful content. However, the approach is reactive in nature and does not provide an effective solution to the problem of preventing the spread of misinformation.

The application of the algorithm auditing approach addresses the root cause of the problem and provides an effective solution. The approach provides long-term solutions to the problem and can be highly effective in preventing the spread of misinformation.
Figure 2 provides an overview of the application of a system like this.

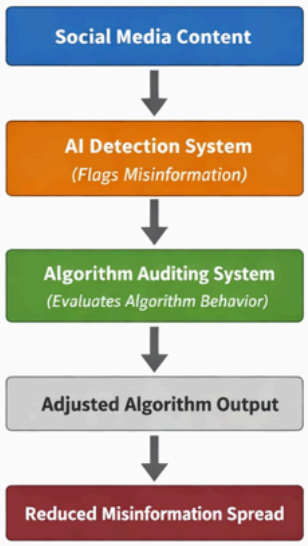

Figure 2: *Integrated AI Detection and Algorithm Auditing Framework*

The application of this approach provides immediate and long-term solutions to the problem. The application of AI detection systems provides immediate solutions to the problem, whereas the application of the algorithm auditing approach provides long-term solutions.
However, there are several issues that need to be taken into consideration. First and foremost, the application of this approach requires the collaboration of technology companies and researchers. In addition to this, the application of this approach needs to be carefully analyzed in the context of the ethical issues that it may raise.

The application of the AI detection approach in conjunction with the algorithm auditing approach provides a feasible and scalable engineering solution. The application of these two approaches in conjunction with each other has the potential to provide an effective solution to the problem.

6. Conclusion

The dissemination of misinformation on social media sites is an important problem affecting current digital communication infrastructures. Artificial intelligence and algorithm auditing technologies are important tools for fighting the dissemination of misinformation on social media sites.

Despite the fact that artificial intelligence and algorithm auditing technologies are not the best solutions to the dissemination of misinformation on social media sites, further research and development on the technologies can go a long way in reducing the effects of misinformation on social media sites and other digital communication infrastructures. Through technological innovation and good governance of social media sites, a reliable and trustworthy digital environment can be achieved.

7. References

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Appendix:

The feedback I got from my peers for my original draft was that i didnt correctly use in text citation to make it clear whose idea I was portraying in my report. My table of contents was poorly constructed and I had a lack of visual clarity in terms of explaining the content I was talking about. Everything about my original report lacked clarity and expansion.

Now after taking all that feedback into consideration i have created a well refined report and critical analysis of my data while giving the reader a point of view different to most on Large Language Models.

AI Disclosure:

AI was used to clarify terms used by the various authors I have cited to make it easier for me to understand what the main ideas are, some of which were used in the creation of this document. It was also used to give me critical feedback on where i lacked in terms of the grading rubric.